Windows Transaction Load

Revision 5.5

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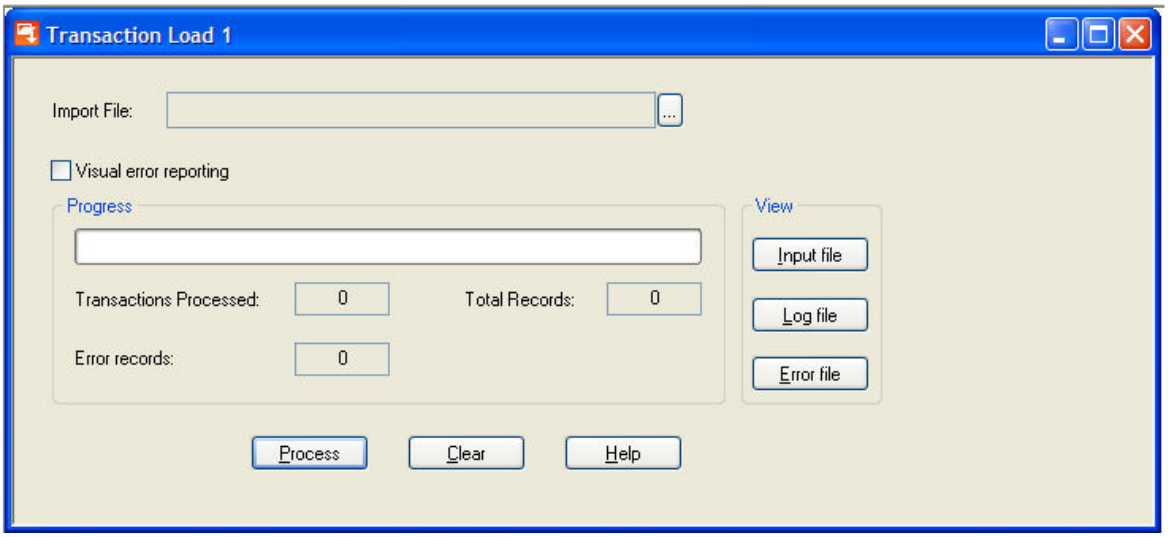
1. Transaction Load

Windows Transaction Load is a MAX program (WTL.exe) used to load transactions into MAX.



When WTL.exe opens, the following dialog box displays.

To load transaction data into MAX, all transactions must be included in a single, ASCII file formatted using the information that follows.

1.1 Rules and Validation

You cannot load inventory transactions for parts with a Part Type Code of F (Family). Because they are non-unique parts, Family parts cannot be stocked.

Set the Use Tnx Date/Time switch in System Configuration to Yes if you want to load user input dates. Set it to No if you want the transactions dated with the current system date.

Enter or scroll and choose the file name to be loaded. You can choose to view the input file before loading, view the error log which shows records that were not loaded, or view the log for all transactions.

This option loads <u>All</u> transactions contained in the specified file provided that they are stored in the required format as listed below. Also, there must be sufficient inventory on hand in the specified location for issue transactions (A, F, S and I transaction types). In other words, transactions that will take the balance of a part below 0 in a part stock location will be disallowed.

If any of your product structures were entered with a Negative Quantity Per Assembly, when processing a receipt to stock using this option the system will also receive the by-product of that manufacturing process. The system calculates the quantity of the by-product being received using the following calculation:

(Quantity Per Assembly * -1) * (Quantity Received To Stock)

If you have the Vouchered Receipt flag set to Y in Accounting Integration configuration, Purchase Order Receipt (PORECS) records will be created for all receipt (R) transactions of sub type N (Non-Inventory Purchase Orders) or P (Inventory Purchase Orders).

The system automatically edits certain fields to ensure that any changes made will be valid within the system. For example, this option verifies that the Transaction Type is equal to **A**, **B**, **C**, **F**, **I**, **P**, **R**, **S**, **T**, **W** or **Y**. Any other values will cause an error message to be sent to the error log and that transaction will be rejected.

System input files are either Alphanumeric (letters and numbers), Integers (whole numbers), or numbers with Decimal places. These are abbreviated as A, I, and D.

Note: When a field is specified as D, the field length includes the decimal point and the numbers to the right of the decimal point (i.e., if a field length is 10 with 2 decimal places, the largest number is 9999999.99). Field lengths for type D are specified as 10(2) meaning 10 total spaces with 2 decimal places.

1.2 File Renaming

When the system loads the ASCII input file *.INP if errors are encountered, an error log and rejected record files is created with the same input file name, but different file extensions of .ERR and .FIX respectively. This way the input, reject and error log files will be a matched set. Also, to prevent the same input file from being loaded multiple times, the input file extension is changed to *.INX.

If for some reason the system cannot read one or all of the transactions in the file specified, error messages will be sent to an error log informing you of the transaction involved and the reason for the rejection. The error log may be viewed or printed. Upon completion of the load function, a message appears telling you how many records were successfully loaded.

The table below shows the field names, the length of the field, the field type, and the position of each field. The input file must be in this format for a successful load.

1.3 MAXLog Transaction Load Format

Description	Type	Length
Transaction Type	A	1
Subtype	A	1
Receive to Stock Id	A	8
Issue from Stock Id	A	8
Part Id	A	30
Order Number	A	8
Line Number	A	2
Delivery Number	A	2
Transaction Qty	N	10
Transaction date	A	6
Transaction time	A	6
Reference description	A	20
GL Reference	A	3
Operation sequence number	A	4

Description	Type	Length
Next operation sequence number	A	4
Actual run time	N	10
Actual setup time	N	10
Shift	I	1
Zone (to Stock ID)	A	10
Employee id	A	7
Ticket number	A	6
Start time	A	4
End time	A	4
Setup time	N	10
Elapsed time	N	10
Time ticket scrap qty	N	10
Reason code	A	1
User name	A	100
User defined key	A	15
User defined reference	A	25
Assignment code for serial numbers (R, I)	A	1
Lot number	A	30
Serial number	A	30
Terminal Id	A	3
Quarantine	A	1
Expiration date	A	6
Defect	A	4
RECPL	A	1
DISPOSITION	A	15

	TYPE OF TRANSACTION
A	Adjustment
B	Post Production Completion
C	Cycle Counts
F	Stock Transfer From
I	Issue From Stock
P	Post Operation (Subtype Required)
R	Receipt to Stock (Subtype Required)
S	Shipment
T	Time Ticket data
W	Rework
Y	Scrap

	SUBTYPE OF TRANSACTION
B	Backflush
C	Completion (Partial or Complete)
N	Non Inventory
O	Sales Order (Subtype of Shipment)
P	Purchase Order
S	Shop Order
T	Time Ticket Completion
U	Unplanned

Key

A	Alphanumeric
N	Numeric Only
D	Date (Year 2000 is 00)
10	Ten total places including decimal point
HHMM	HourHourMinuteMinute in military format

1.4 Validation Rules

****RECEIVE TO STOCK ID*** ***A*** ***8***

Used in Receipt to stock and transfer stock to stock transactions. Must be a valid stock location in the stock master file. If this field is blank and required for a particular transaction, the primary stock location for the part being transacted will be used.

****ISSUE/TRANSFER FROM STOCK ID*** ***A*** ***8***

Must be a valid stock location with enough qty on hand in that location to perform the issue/transfer. If this field is left blank, the PRIMARY STOCK LOCATION for the part ID being loaded will be used.

****PART ID*** ***A*** ***30***

Must be valid if required by operation type / subtype. Non inventory transactions (shipments) allow any part number.

ORDER #** ***A ***8***

If required by transaction type and subtype.

****LINE #*** ***A*** ***2***

If required by transaction type and subtype.

****DELIVERY #*** ***A*** ***2***

If required by transaction type and subtype.

****TRANSACTION QTY*** ***N*** ***10***

Any number between -999999 and 999999 depending on the type of transaction. Negative receipts to stock are allowed. MAXLog allows 3 decimals10,3.

TRANSACTION DATE ***D*** ***6***

Will use the value contained in this field if not empty and is a valid date. If this field is empty, it will use the current system date. System Configuration > System Switches > Trans Date Time must be set to “Y” in order for the Transaction Load module to use the date in the input file records.

TRANSACTION TIME ***A*** ***6***

Will use the value contained in this field if not empty and is a valid time based on the 24 hour clock. I.E. 175000 is 5:50:00 pm. If this field is empty, it will use the current system time. System Configuration > System Switches > Trans Date Time must be set to “Y” in order for the Transaction Load module to use the time the input file records.

REFERENCE DESCRIPTION ***A*** ***20***

User defined

****GL REFERENCE*** ***A*** ***3***

3-character override must be valid in Maintain GL override accounts.

OPERATION SEQUENCE #** — ***A — ***4***
If required by transaction type and subtype. Must be a valid operation sequence for the order specified.

NEXT OPERATION SEQUENCE #** — ***A — ***4***
If required by transaction type and subtype. Must be a valid operation sequence for the order specified.

ACTUAL RUN TIME — ***N*** — ***10***

ACTUAL SETUP TIME — ***N*** — ***10***

SHIFT — ***I*** — ***1***
1, 2, or 3

ZONE — ***A*** — ***10***
User defined will pull from prtmst->zone is blank and prtmst is not empty.

EMPLOYEE ID** — ***A — ***7***
Must be valid employee id (if required by transaction type).

TICKET # — ***A*** — ***6***

START TIME — ***hhmm*** — ***A*** — ***4***

END TIME — ***hhmm*** — ***A*** — ***4***

SETUP TIME — ***N*** — ***10***

ELAPSED TIME — ***N*** — ***10***

TIME TICKET ACTUAL SCRAP QTY — ***N*** — ***10***

REASON CODE — ***A*** — ***1***

USER NAME** — ***A — ***100***
Must be a valid User Name.

USER DEFINED KEY — ***A*** — ***15***

USER DEFINED REFERENCE — ***A*** — ***25***

ASSIGNMENT CODE FOR SERIAL #'S — ***A*** — ***1***
Should be I or R if part is under serial control, blank otherwise.

LOT NUMBER** — ***A — ***30***

***SERIAL NUMBER** *A* *30*

***TERMINAL ID** *A* *3*
Automatically assigned by the terminal.

***QUARANTINE** *A* *1*
Yes or No.

***EXPIRATION DATE** *D* *6*
Expiration date of the Lot. If this field is left blank and there is a Shelf Life on the Part Master record, the Expiration Date will automatically be calculated after the Transaction Load process is complete.

CASE
Data is case sensitive. For example, if you have a Stock ID of "FG" in the system, you can not load this stockroom as "fg" in the input file.

1.5 MAXLog Required Fields Matrix

Transaction Type	A	B	C	F	I	I	P	P	R
Transaction Sub Type					S	U	C	T	B
Transaction Date	✓	✓	✓	✓	✓	✓	✓	✓	✓
Transaction Time	✓	✓	✓	✓	✓	✓	✓	✓	✓
Order Number					✓		✓	✓	✓
Line Number									
Delivery Number									
Operation Sequence							✓	✓	
Next Operation Sequence							O		
Part Identifier	✓	✓	✓	✓	✓	✓			
Issue From Stock ID	✓	O	✓	✓	✓	✓			
Receive to Stock ID		✓		✓					O
Locator (bin #)	O		O	O					O
Transaction Quantity	✓	✓	✓	✓	✓	✓	✓	✓	✓
Reference Description	O	O	O	O	O	O	O		O
GL Acct. Override	O		O	O	O	O			O
Shift							✓	O	
Employee ID								✓	
Ticket Number								✓	
Start Time								✓	
End Time								O	
Setup Time								O	
Elapsed Time								O	
Actual Run Time							✓	✓	
Actual Setup Time							✓	✓	
Time Ticket Scrap Qty								✓	
Reason Code								O	
User Name	O	O	O	O	O	O	O	O	O
User Defined Key	O	O	O	O	O	O	O	O	O
User Defined Reference	O	O	O	O	O	O	O	O	O
Assignment Code for Serial #'s	✓		✓	✓	✓	✓			✓
Lot Number	O		O	O	O	O			O
Serial Number	O		O	O	O	O			O
Terminal ID	✓	✓	✓	✓	✓	✓	✓	✓	✓
Quarantine	O		O	O	O	O			O
Expiration Date	O		O	O	O	O			O

MAXLog Required Fields Matrix (Cont'd)

Transaction Type	R	R	R	R	S	S	T	W	Y
Transaction Sub Type	N	P	S	U		O			
Transaction Date	✓	✓	✓	✓	✓	✓	✓	✓	✓
Transaction Time	✓	✓	✓	✓	✓	✓	✓	✓	✓
Order Number	✓	✓	✓		✓	✓	✓	✓	✓
Line Number	✓	✓			✓	✓		✓	
Delivery Number		✓			✓	✓		✓	
Operation Sequence							✓	✓	✓
Next Operation Sequence								O	
Part Identifier				✓					
Issue From Stock ID					✓	✓			
Receive to Stock ID		O	O	✓					
Locator (bin #)	O	O	O	O					
Transaction Quantity	✓	✓	✓	✓	✓	✓	✓	✓	✓
Reference Description		O	O	O	O	O		O	O
GL Acct. Override	O	O	O	O	O	O		O	O
Shift							✓		O
Employee ID							✓		
Ticket Number							✓		
Start Time							✓		
End Time							✓		
Setup Time							O		
Elapsed Time							O		
Actual Run Time							✓		
Actual Setup Time							✓		
Time Ticket Scrap Qty							✓		
Reason Code									
User Name	O	O	O	O	O	O			O
User Defined Key	O	O	O	O	O	O		O	O
User Defined Reference	O	O	O	O	O	O		O	O
Assignment Code for Serial #'s		✓	✓	✓	✓	✓			
Lot Number		O	O	O	O	O			
Serial Number		O	O	O	O	O			
Terminal ID	✓	✓	✓	✓	✓	✓	✓	✓	✓
Quarantine		O	O	O	O	O			
Expiration Date		O	O	O	O	O			

Legend

✓	=	Required Field
O	=	Optional Field
Blank	=	Not Used

1.6 Rework Transactions

Rework transactions are somewhat unique. This function gives you the ability to reset the load and queue quantities for an order, and send the job back to an operation already completed because of some quality or performance problem.

You need only identify the order number, quantity and operation sequence for this input file. All of the other information already exists in the system.

The rework function will set the queue quantity at the specified operation to the sum of the parts completed at that operation plus all subsequent operations. The quantity completed at the specified operation, and all subsequent operations, will be reset to zero.

The total load, total queue, total input and smoothed input information on each workcenter involved will also be reset to accurately reflect the status of the job.

In other words, everything in the job that has been completed at or beyond the specified operation will all be sent back to the operation that needs to be reworked.

There will be no transaction history record created as a result of this function.

Because of the nature of this function, we recommend that you only specify one operation per order number. If there are multiple operations that all need to be reworked, specify the EARLIEST operation; all subsequent operations have to be redone anyway.

1.7 Lot and Serial Number Transactions

Serial number assignments are handled through three methodologies: individual, range or none. A field Assignment Code added for Lot/Serial Number assignment which will accept three entries that correspond to these: I, R, and N. (N is not applicable for parts under Lot only.) The user may decline to assign serial numbers at the time of receipt (Assignment Code = N) and can assign them at a later time by either manual assignment (Cycle Count) or automatic assignment (Update Serialized Receipts). The individual Serial Master records will be created for each unit with an asterisk “*” in the serial number field. A range of serial numbers can be assigned or selected (Assignment Code = R) by entering the first serial number and the system will assign the next sequential range of serial numbers equal to the transaction quantity (only numeric serial numbers can be accommodated by this method). The individual assignment (Assignment Code = I) is handled through the ASCII file where individual serial numbers can be either selected or assigned to a transaction, with validation of transaction quantity = the quantity of assigned numbers. The user must specify the assignment code of **I** and a record is created for the entire transaction quantity.

As an example for unplanned or shop receipt: Transaction Load will read the first record as individual assignment, create one transaction history record for a quantity of 15 then create and validate a serial number record for the next 14 entries.

Transaction Load will read these multiple records and put them into a temporary Btrieve file and sort them by Terminal ID and Date/Time, so the records can be stored and processed later in the sequential order that they were collected on that terminal. Receipts or issues of multiple Lot Numbers would be transacted individually (i.e. one receipt transaction for each Lot number within an order), as is currently done in the system. Items under Lot and Serial control will conform to the rules of Lot numbers first, then serial numbers. In a practical sense thus receipts are processed before shipments for the same part id.

There is a Lot Tracking transaction subtype for Cycle counts of 'L'. If the subtype is 'L' it works similarly to the individual serial number option above. The first line has the Cycle type and Subtype "L" with the total cycle count quantity (but no lot number). The next line(s) have the lots to cycle count with a quantity for each. The quantities of each of the individual lots must add up to the total quantity on the first line.

1.8 Repetitive Transactions

Repetitive transactions do not allow the parent or any of the component parts to be under Lot or Serial tracking control.

1.9 Cycle Counts under Lot Control

Each line for a cycle count for a lot part must begin with CL. The first required line of the file is a dummy line with the overall cycle count quantity. The second line contains the lot number and the lot quantity. The third and following lines contain lots number and lot quantity the same as the second until the total cycle count quantity on line one is used up. If the individual lot quantities do not add up to the total cycle count on line one, transaction load will write an error message to the *.ERR file.

When Cycle Counting Lots or Individual Serial Numbers the combination of Terminal ID Date and Time must be unique on each separate transaction like the following example:

CYCLE COUNT PART 1000 QTY 24
LOT/Qty/ TermID/Date/Time
1 ...12 001/063103/080101
2.... 6 001/063103/080101
3 ....6 001/063103/080101

Next Lot Time Goes Forward

CYCLE COUNT PART 1100 QTY 3
LOT/Qty/ TermID/Date/Time
1 ...1 001/063103/080102
2....1 001/063103/080102
3 ...1 001/063103/080102

These 3 elements are used as internal pointers for sorting when adding the many lots into the MAX data base against the cycle count qty.